MIL-C-81777B(AS)
30 November 1973
Superseding
MIL-C-81777A(AS)
30 November 1972

MILITARY SPECIFICATION
COOLER, LIQUID, ELECTRONIC EQUIPMENT
HD-908 ( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. The equipment covered by this specification shall be designated as the Electronic Equipment Shop Liquid Cooler, herein referred to as the LC. The LC shall be for intermediate level use to condition the liquid coolant supplied to test stations during checkout and test of certain weapon replaceable assemblies (WRA's) from the radar subsystem of the AN/AWG-9 Airborne Weapon Control System (AWCS).

1.2 Classification. The LC (493701-150) shall consist of a cabinet assembly and a remote switching control. The cabinet assembly shall contain a coolant circulation loop, a refrigeration system, local controls and indicators, and electronic subassemblies.

a. Cabinet assembly, major elements

Drawing or Model Number	Nomenclature	Reference Paragraph
251177	Condenser, Water	3.6.1
251178	Receiver, Refrigeration	3.6.2
251180	Chiller	3.6.3
251190	Regulator, Temperature	3.6.4
251191	Compressor, Reciprocating	3.6.5
251192	Pumping Unit - Hydraulic	3.6.6

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Drawing or Model Number	Nomenclature	Reference Paragraph
251200	Valve-Thermostatic Expansion	3.6.7
3069032	Panel, Assembly, Control and Self-Check	3.6.9
3069038	Starter Assembly, Motor	3.6.10
3069081	Enclosure Assembly, Relay	3.6.11

b. Remote switching control

Drawing or Model Number	Nomenclature	Reference Paragraph
3069031	Control, Remote Switching	3.6.8

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in the contract, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-R-16743	Refrigerating Plants and Systems, Mechanical, and Refrigerating System Components - Dichlorodifluoromethane Type 12 -- Naval Shipboard
MIL-E-17555	Electronic and Electrical Equipment Accessories, and Repair Parts; Packaging and Packing of
MIL-T-81734 (AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

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STANDARDS

Military

MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-761	Electric Power, Alternating Current for Shipboard Use, Characteristics and Utilization of

DRAWINGS

Hughes Aircraft Company

251177	Condenser, Water
251178	Receiver, Refrigeration
251180	Chiller
251190	Regulator, Temperature
251191	Compressor, Reciprocating
251192	Pumping Unit, Hydraulic
251200	Valve-Thermostatic Expansion
3069031	Control, Remote Switching
3069032	Panel, Assembly, Control and Self-Check
3069038	Starter Assembly, Motor
3069081	Enclosure Assembly, Relay
493150-150	Test Station, Radio Frequency
493350-150	Test Station, Low Frequency
493701-150	Cooler, Liquid, Electronic Equipment Shop

2.2 Availability of documents. When requesting specifications, standards, drawings and publications, refer to both title and number. Copies of applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Pennsylvania 19120.

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3. REQUIREMENTS

3.1 Materials, parts, and processes. Materials, parts, and processes shall be in accordance with MIL-T-81734.

3.2 Design and construction. Design and construction shall be in accordance with MIL-T-81734 and Drawing 493701-150.

3.2.1 Design features. The design features of MIL-T-81734 shall apply with the following additions.

3.2.1.1 Size and weight. The cooling stand shall not exceed 48 inches by 36 inches by 72 inches. The remote control panel shall not exceed 12 inches by 6 inches by 18 inches. Loading, averaged over the total surface area for the cooling stand, shall not exceed 250 pounds per square foot. The weight of the cooling stand, including fluids, shall not exceed 3000 pounds.

3.2.2 Shop facilities. Shop facilities will be in accordance with MIL-T-81734.

3.2.2.1 Electrical power. Facility supplied electrical power forms shall be in accordance with MIL-T-81734. Alternating current (ac) power forms shall be in accordance with MIL-STD-761, Type I. The electrical power required for operation of the LC shall be not more than 20.0 kilovolt amperes(kVa) of 440 volt (v), 60 Hertz (Hz), 3-phase, 3-wire, ungrounded.

3.2.2.2 Cooling water. Cooling water will be supplied in accordance with MIL-T-81734.

3.2.2.3 Liquid coolant. The liquid coolant conditioned by the LC shall be Coolanol 25 (Monsanto Chemical Company), Oronite Flo-Cool 180 (Chevron Chemical Company), or equivalent.

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3.2.2.4 Nitrogen. Nitrogen will be provided in accordance with MIL-T-81734.

3.2.2.5 Freon. Freon, Type 12, will be provided by the facility.

3.2.3 Service conditions

3.2.3.1 Electrical. The LC shall meet the performance requirements of this specification when supplied with the electrical power specified in 3.2.2.1.

3.2.3.2 Environmental. Environmental service conditions shall be in accordance with MIL-T-81734.

3.2.4 Standard test conditions. Standard test conditions shall be in accordance with MIL-T-81734.

3.2.5 Maintainability. Maintainability shall be in accordance with MIL-T-81734.

3.2.5.1 LC self test. The LC shall provide capability to establish functional readiness of the liquid cooler. Time to perform self test shall not exceed 30 minutes. The self test shall include capability to override the LC control of the remote switching control (3069031).

3.2.5.2 LC mean time to repair (MTTR). The LC shall provide capability to isolate faults to the shop replaceable items. The time required for fault isolation, replacement, adjustments, and repair verification shall not exceed a mean time of 4 hours for any single malfunction.

3.2.5.3 Test points. Test points shall be in accordance with MIL-T-81734. Test points shall be provided to facilitate fault isolation to the shop replaceable item, fault isolation within the shop replaceable item, for alignments, adjustments, calibration, and signal monitoring. Location and accessibility of test points shall be dictated by considerations of ease of maintenance and repair.

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3.2.5.4 Calibration cycle. The LC calibration cycle shall be in accordance with MIL-T-81734.

3.2.6 Reliability. The LC mean-time-between-failure (MTBF) shall be 500 hours, minimum.

3.2.6.1 Operating life. Operating life shall be in accordance with MIL-T-81734.

3.2.6.2 Storage life. Storage life shall be in accordance with MIL-T-81734.

3.2.6.3 Operational stability. Operational stability shall be in accordance with MIL-T-81734.

3.2.7 Electromagnetic interference. The limits of electromagnetic interference for the liquid cooler shall be in accordance with MIL-T-81734 and the following:

a. MIL-STD-461 tests CE01 and CE02 and the limits therein shall apply.

b. MIL-STD-461 tests CE03, CE04, CS01, CS02, CS06, RE02, RS01, RS02, RS03 and RS04 shall not apply.

3.2.8 Power switching and coolant control. The LC shall provide the necessary switching and hydraulic devices to control facility supplied power forms and liquid coolants.

3.2.9 Emergency shutdown switch. An emergency shutdown switch shall be provided for use by the operator. When the emergency shutdown switch is activated, the load side of all LC primary circuit breakers shall be disconnected from their respective loads. The switch utilized for this application shall be of the pushbutton activated, momentary type, with the pushbutton finished in red. The front panel area around the pushbutton shall be marked with alternating orange-yellow and black stripes. The switch shall be centrally located on the LC in an area which is easily accessible to the operator during normal operation.

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3.2.10 Displays. Displays provided on the cabinet assembly and on the remote switching control shall provide an indication of LC operational status.

3.2.11 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the LC and shall be capable of registering 9,999 hours.

3.2.12 Installation. The LC shall be capable of supplying coolant to the radio frequency test station (RFTS) and the low frequency test station (LFTS) as follows:

a. A total of 50 psi dropped in the supply and return lines provided by the shop facility

b. A maximum vertical head not to exceed 40 feet above or below the test stations.

3.2.13 LC interface. The LC shall provide the interfaces specified below.

3.2.13.1 Shop liquid cooler/test station interfaces. The LC shall interface with the RFTS (493150-150) and LFTS (493350-150) through two facility coolant lines (supply and return) which enter the LC.

3.2.13.2 Shop liquid cooler/cooling water interface. The LC shall interface with the shop cooling water through two facility water lines (supply and return) which enter the LC.

3.2.13.3 Shop liquid cooler/shop electrical power interface. The shop electrical power specified in 3.2.2.1 shall interface with the LC through facility power lines which enter the LC.

3.2.14 Interchangeability. Interchangeability shall be in accordance with MIL-T-81734.

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3.3 Performance. The LC shall provide liquid cooling with the following parameters:

a. Supply pressure. Pressure of the coolant at the output of the LC shall be adjustable between 185 pounds per square inch gage (psig) and 130 psig and shall be accurate to ±10 psig of the setting. Pressure shall be maintained for any flow up to and including 10 gallons per minute (gpm).

b. Total heat load. The LC shall have the capacity to remove a maximum heat load of 102,400 British thermal units per hour (Btu/hour).

c. Coolant outlet temperature. Coolant outlet temperature shall be $+18.3^{+11}_{-8.3}$ °C ($+65^{+20}_{-15}$ °F).

d. Coolant filter. 10 microns, absolute.

3.4 Detailed functional requirements. The LC shall consist of a coolant circulation loop, a refrigeration system, and local/remote controls and indicators.

3.4.1 Coolant circulation loop. Reservoir-stored coolant shall be filtered and then pumped to the refrigeration system heat exchanger for cooling, or directly to the mixing valve. Coolant shall be mixed to attain the proper output temperature and then transferred to the test station. Temperature and pressure gages shall provide for monitoring of output coolant pressure and temperature. A bleed valve shall be provided at the high point and a drain valve shall be provided at the low point.

3.4.1.1 Coolant filter. A full-flow, depth type disposable filter shall be located in the coolant supply line. The filter shall have an absolute removal rating of 10 microns and shall include a differential pressure indicator to provide an indication of dirt accumulation.

3.4.1.2 Coolant reservoir. A coolant reservoir having a capacity of at least 10 gallons and pressurized by nitrogen at a pressure equal to or less than 20 psig shall be provided. The reservoir shall include a safety relief valve to prevent pressure over 25.0 ±2.5 psig. The pressurized nitrogen shall be supplied from a nitrogen cylinder located within the cooling stand.

3.4.2 Refrigeration system. A refrigeration system, in accordance with MIL-R-16743, shall be provided for chilling the coolant. Fresh or sea water shall be circulated through the refrigeration system

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condenser for dissipation of heat. Gages shall be provided for monitoring the refrigerant.

3.4.3 Controls and indicators. The controls and indicators shall be provided locally and remotely. The controls and indicators shall provide monitoring and control of the LC functions.

3.5 Workmanship. Workmanship shall conform to MIL-STD-454, Requirement 9.

3.6 LC elements. The LC shall contain the following items, according to the configuration of 493701-150.

3.6.1 Condenser, water. The water condenser shall cool and condense high pressure gas from the compressor into liquid. The condenser shall be cooled by fresh or sea water. The maximum water flow through the condenser shall be 30 gpm.

3.6.2 Receiver, refrigeration. The refrigeration receiver shall store liquid refrigerant on the high pressure side of the refrigeration system. The receiver shall have a capacity of 92 pounds of Freon 12 (based on 80 percent full capacity at 86° Fahrenheit (F).

3.6.3 Chiller. The chiller shall act as a heat exchanger between the refrigeration loop and the liquid coolant loop.

3.6.4 Regulator, temperature. The temperature regulator shall control and regulate the temperature of the coolant prior to the coolant being pumped to the test stands. The temperature range of the regulator shall be 50 to 90°F.

3.6.5 Compressor, reciprocating. The reciprocating compressor shall receive the low pressure vaporized refrigerant, compress it, then discharge hot high pressure gas to the condenser. The induction motor employed shall operate from 440 Vac, 3-phase, delta, 60 Hz, or 200/220 Vac, 3-phase, 60 Hz input. The compressor displacement shall be 2380 cubic feet per hour.

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3.6.6 Pumping unit-hydraulic. The pumping unit shall deliver reservoir-stored coolant to the chiller, or directly to the test stands through the temperature regulator mixing valve. The pumping unit motor shall operate from 220/440 Vac, 3-phase, 60 Hz power. The maximum pump capacity shall be 13 gpm at 1500 psig pressure.

3.6.7 Valve-thermostatic expansion. The thermostatic expansion valve shall regulate the flow of refrigerant to the chiller. The valve capacity shall be 11 tons refrigeration and shall have a thermostatic charge suction temperature range of 0 to 35°F.

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3.6.8 Control, remote switching. The remote switching control (3069031) shall provide control and monitoring of the cooling stand when interconnected by an 80-foot electrical cable. Coolant monitoring and malfunction indications shall be included. The following controls and indicators shall be located on the panel:

a. Supply pressure meter (0-250 psi)

b. Supply temperature meter (40 to 140°F)

c. Return temperature meter (40 to 140°F)

d. Low water flow indicator light

e. Low reservoir level indicator light

f. Low reservoir pressure indicator light

g. High coolant temperature indicator light

h. Low coolant temperature indicator light

i. Low coolant flow indicator light

j. Remote control override indicator light

k. Power on indicator light

l. Power on switch

m. Power off switch

n. Lamp test switch

o. Cooler ready indicator light

3.6.9 Panel, assembly, control and self check. The panel control and self check assembly (3069032) shall provide control and monitoring of the LC. Monitoring and malfunction indications for the coolant and compressor shall be included and located on the panel in a manner accessible from the front of the LC cabinet with the cover removed.

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3.6.9.1 Indicators and controls. The indicators and controls mounted on the control and self check panel assembly shall be as follows:

a. Low oil indicator light

b. Over temperature indication light

c. Head pressure gage (0-300 pounds per square inch (psi))

d. Suction pressure gage (30 inches of vacuum to 300 psi)

e. Oil pressure gage (0-300 psi)

f. Low flow indicator light

g. Supply over temperature indicator light

h. Supply low temperature indicator light

i. Supply temperature gage (-40 to +130°F)

j. Return temperature gage (40 to 240°F)

k. Supply pressure gage (0-300 psi)

l. Return pressure gage (0-100 psi)

m. Return high temperature indicator light

n. Reservoir low level indicator light

o. Reservoir low pressure indicator light

p. Power on indicators

q. Coolant pump switch

r. Power on switch

s. Compressor switch

t. Remote control override switch

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u. Water low flow indicator light

v. Manual by-pass switch

w. Lamp test switch

x. Emergency off switch

y. Elapsed time indicator

3.6.10 Starter assembly, motor. The motor starter assembly (3069038) shall provide the electrical circuits to distribute and control power to the LC. A circuit breaker and an emergency off switch shall be located on the assembly. The emergency off switch shall be accessible from the front of the LC cabinet with the cover panel on or off. Test points for electrical power forms shall be provided and located inside the assembly.

3.6.11 Enclosure assembly, relay. The relay enclosure assembly (3069081) shall provide electrical circuits for sensing malfunctions within the LC.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for tests and inspections. Responsibility for tests and inspections shall be in accordance with MIL-T-81734.

4.2 Tests. Tests shall be conducted in accordance with MIL-T-81734.

4.3 Reliability verification. Verification of the reliability requirements shall be in accordance with MIL-T-81734.

5. PREPARATION FOR DELIVERY

5.1 General. Preparation for delivery shall be in accordance with MIL-E-17555.

6. NOTES

6.1 Ordering data. Ordering data shall be in accordance with MIL-T-81734.

6.2 Definitions. Definitions shall be in accordance with MIL-T-81734.

6.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract. The contract shall have precedence over any other document.

b. This specification. This specification shall have precedence over all applicable subsidiary documents.

c. Reference documents. Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

Project No. 4935-N103

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No, 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-81777B(AS) COOLER, LIQUID, ELECTRONIC EQUIPMENT HD-908( ) /AWM-23(V)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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